Addis Ababa Science & Technology University

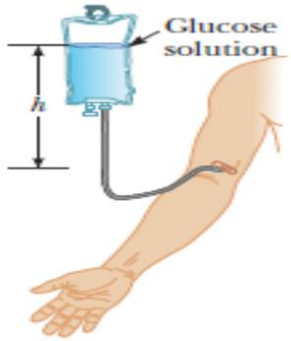
College of Natural & Applied Sciences

Department of Physics & Statistics

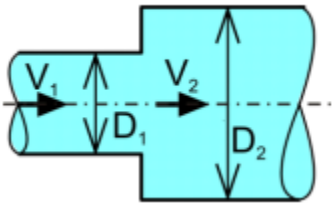
Worksheet 3 for General Physics (Phy1001)

1. A 66 kg man lies on his back on a bed of nails with 1208 of the nails in contact with his body. What is the total pressure exerted on the man's body if the tip of each nail has an area of 10^{-6} m^2

2. Glucose solution of density 1025 kg/m^3 is given to a patient from a collapsible plastic bag as shown in the figure below. If the average gauge pressure in the vein is 1.23×10^3 Pa, what must be the minimum height h of the bag to infuse glucose into the vein? $\left(Take\ g = 10\frac{m}{s^2}\right)$

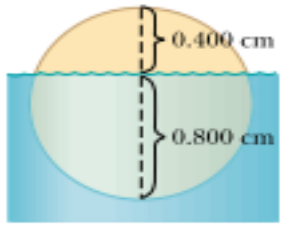


3. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 5.00 cm. This pressure is transmitted by a liquid to a piston that has a radius of 15.0 cm. What force must the compressed air exert to lift a car weighing 13 300 N? What air pressure produces this force?
4. On a circular conduit there are different diameters: diameter D_1 = 2 m changes into D_2 = 3 m. The velocity in the entrance profile was measured: V_1 = 3 m/s. Calculate the discharge and mean velocity at the outlet profile (see fig below).

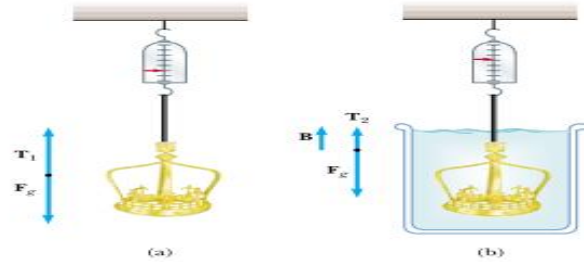


5. A liquid flows through a pipe with a diameter of 10cm at a velocity of 9cm/s. If the diameter of the pipe then decreases to 6cm, what is the new velocity of the liquid?
6. A pipe with a diameter of 4 centimeters is attached to a garden hose with a nozzle. If the velocity of flow in the pipe is 2m/s, what is the velocity of the flow at the nozzle when it is adjusted to have a diameter of 8 millimeters?

7. Water flows through a fire hose of diameter 6.35 cm at a rate of 0.0120 m^3/s. The fire hose ends in a nozzle of inner diameter 2.20 cm. What is the speed with which the water exits the nozzle?
8. A wooden dowel has a diameter of 1.20 cm. It floats in water with 0.400 cm of its diameter above water (Fig below). Determine the density of the dowel.



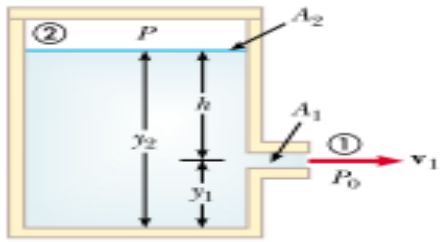
9. A cube of wood having an edge dimension of 20.0 cm and a density of 650 kg/m3 floats on water. (a) What is the distance from the horizontal top surface of the cube to the water level? (b) How much lead weight has to be placed on top of the cube so that its top is just level with the water?
10. A piece of wood with density $\delta = 0.6$ g/cm^3 floats on the surface of water. Find the ratio of the portion of the wood submerged in the water to that of outside the water? (give density of water = 1g/cm^3)
11. An object with a mass of 150 kg is thrown into the water and displaces a volume of 100lit of water. Will the object sinks or floats?
12. What is the buoyant force on a raft with a volume of 6 cubic meters which 2/3 of its volume is submerged in sea-water of density 1025 kg/m^3
13. Archimedes supposedly was asked to determine whether a crown made for the king consisted of pure gold. Legend has it that he solved this problem by weighing the crown first in air and then in water, as shown in Figure below. Suppose the scale read 7.84 N in air and 6.84 N in water. What should Archimedes have told the king?



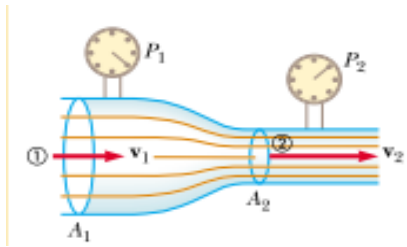
14. A horizontal pipe 10.0 cm in diameter has a smooth reduction to a pipe 5.00 cm in diameter. If the pressure of the water in the larger pipe is 8.00 x10^4Pa and the pressure in the smaller pipe is 6.00 x10^4 Pa, at what rate does water flow through the pipes?

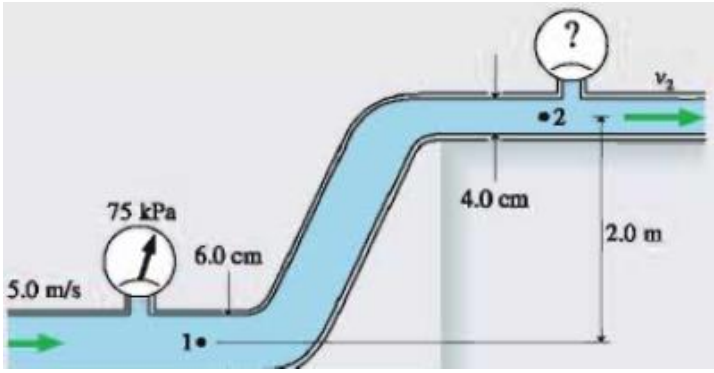
15. A large storage tank, open at the top and filled with water, develops a small hole in its side at a point 16.0 m below the water level. If the rate of flow from the leak is equal to 2.50 x10^{-3} m^3/min, determine (a) the speed at which the water leaves the hole and (b) the diameter of the hole.

16. An enclosed tank containing a liquid of density ρ has a hole in its side at a distance Y_1 from the tank's bottom in the Fig. below. The hole is open to the atmosphere, and its diameter is much smaller than the diameter of the tank. The air above the liquid is maintained at a pressure P. Determine the speed of the liquid as it leaves the hole when the liquid's level is a distance h above the hole.



17. The horizontal constricted pipe illustrated the Fig. below, known as a Venturi tube, can be used to measure the flow speed o incompressible fluid. Determine the flow speed at point 2 if the pressure difference P_1 & P_2 is known.



18. Water circulates throughout a house in a hot water heating system. If the water is pumped at a speed of 0.50 m/s through a 4.0-cm diameter pipe in the basement under a pressure of 3.03 x 105 Pa, what will be the velocity and pressure in a 2.6-cm diameter pipe on the second floor 5.0 m above?

19. Water flow through the pipe in the fig has a diameter of $6cm$ at section (1) and $4cm$ at section (2) . The water's speed through the lower pipe is $5\frac{m}{s}$ and the pressure gauge reads $75kpa$ and point (2) is $2m$ higher than point(1). Calculate the reading of the pressure gauge on the upper pipe?